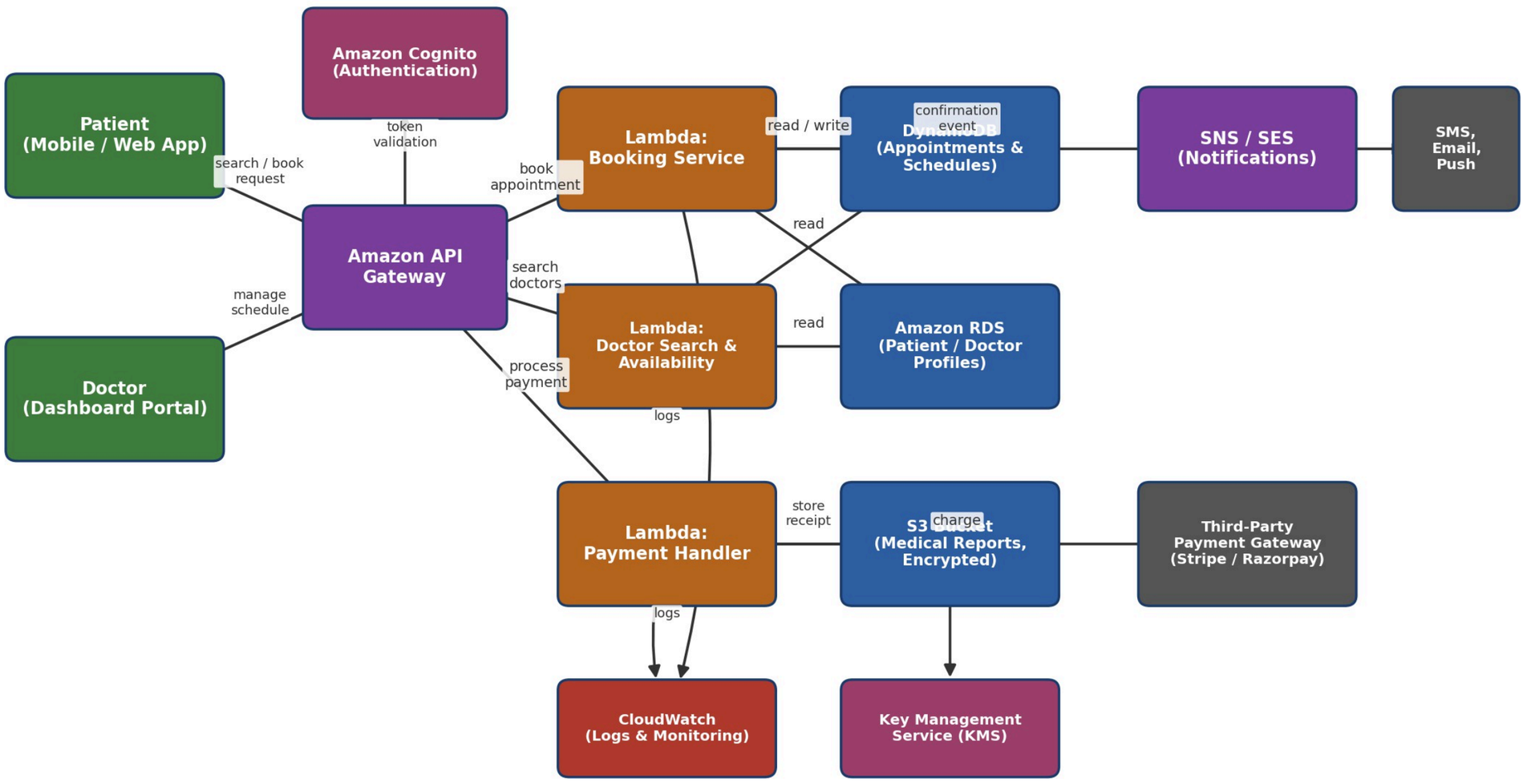
Project Design
Phase Solution
Architecture

Date	4 Aug 2026
Student Name	Navendhiran C
Project Name	Book A Doctor
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.



Book A Doctor – Solution Architecture Diagram:

Figure 1: Architecture and data flow of the Book A Doctor appointment booking application

Architecture Flow Description:

- Patients use a mobile or web app to search for doctors, view availability, and book appointments; doctors use a dashboard portal to manage their schedules.
- Requests pass through Amazon API Gateway, which validates the user's session token with Amazon Cognito before routing the call.
- A Booking Service Lambda handles appointment creation/updates, a Search Lambda finds doctors and checks availability, and a Payment Lambda processes consultation fees.
- Appointment and schedule data is stored in DynamoDB for fast lookups; patient and doctor profile data is stored in Amazon RDS.
- Payment receipts and medical reports are stored in an encrypted S3 bucket, with encryption keys managed by KMS.
- Payments are processed through a third-party gateway (e.g., Stripe/Razorpay), and booking confirmations are pushed via SNS/SES as SMS, email, or push notifications.
- CloudWatch captures logs and metrics from the booking and payment services for monitoring and troubleshooting.

Reference:

https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/